CM_4450_B21___Quarto_Book

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Preface

This is a Quarto book.

To learn more about Quarto books visit https://quarto.org/docs/books.

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1 Point Estimation

1.1 Statistical Inference

- The process of making educated guess and conclusions regarding a population, using a sample of that population is called **Statistical Inference**.
- Two important problems in statistical inference are **estimation of parameters** and **tests of hypothesis**
- Estimation can be of the form of **point estimation** and **interval estimation**.

1.2 Point Estimation

Main Task

- Assume that some characteristic of the elements in a population can be represented by a random variable X.
- Assume that $X_1, X_2, \dots, X_n$ is a random sample from a density $f(x, \theta)$, where the form of the density is known but the parameter θ is unknown.
- The objective is to construct good estimators for θ or its function $\tau(\theta)$ on the basis of the observed sample values $x_1, x_2, \dots, x_n$ of a random sample $X_1, X_2, \dots, X_n$ from $f(x, \theta)$.

Definition: Statistic

Suppose $X_1, X_2, \dots, X_n$ be n observable random variables. Then, a known function $T = g(X_1, X_2, \dots, X_n)$ of observable random variables $X_1, X_2, \dots, X_n$ is called a **statistic**. A statistic is always a random variable.

Definition: Estimator

Suppose $X_1, X_2, \dots, X_n$ is a random sample from a density $f(x, \theta)$ and it is desired to estimate θ. Suppose $T = g(X_1, X_2, \dots, X_n)$ is a *statistic* that can be used to determine and approximate value for θ. Then T is called an **estimator** for θ. An estimator is always a random variable.

Definition: Estimate

Suppose $T = g(X_1, X_2, \dots, X_n)$ be an estimator for θ. Suppose that $x_1, x_2, \dots, x_n$ is a set of observed values of the random variable $X_1, X_2, \dots, X_n$. Then *the value* $t = g(x_1, x_2, \dots, x_n)$ obtained by substituting the observed values in the estimator is called an **estimate** for θ.

- Therefore the **estimator** stands for the function of the sample, and the word **estimate** stands for the realized value of that function.
- *Notation:* An estimator of θ is denoted by $\hat{\theta}$. An estimate of θ is also denoted by $\hat{\theta}$. The difference between the two should be understood based on the context.

Parameter	Estimator: Using random sample $(X_1, X_2, \dots, X_n)$	Estimate 1: Using observed sample $(1, 4, 2, 3, 4)$	Estimate 2: Using observed sample $(4, 2, 2, 6, 3)$
μ	$\hat{\mu} = \bar{X}$	$\hat{\mu} =$	$\hat{\mu} =$
σ^2	$\hat{\sigma^2} = S^2$	$\hat{\sigma^2} =$	$\hat{\sigma^2} =$

1.2.1 Methods of finding point estimators

- In some cases there will be an obvious or natural candidate for a point estimator of a particular parameter.
- For example, the sample mean is a good point estimator of the population mean
- However, in more complicated models we need a methodical way of estimating parameters.
- There are different methods of finding point estimators
  - Method of Moments
  - Maximum Likelihood Estimators (MLE)
  - Method of Least Squares
  - Bayes Estimators
  - The EM Algorithm
- However, these techniques do not carry any guarantees with them.
- The point estimators that they yeild still must be evaluated before their worth is established

1.2.1.1 Method of Moments

- Let $X_1, X_2, \dots X_n$ be a random sample from a population with pdf or pmf $f(x;\theta)$, where $\theta = (\theta_1, \theta_2, \dots, \theta_k)$ and $k \geq 1$.
- Sample moments m' (Sample moments m about 0) and population moments μ' are defined as follows

Sample moment	Population moment
$m_1 = \frac{1}{n}\sum_{i=1}^{n} X_i$	$\mu'_1 = E(X)$
$m_2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2$	$\mu'_2 = E(X^2)$
...	...
$m_k = \frac{1}{n}\sum_{i=1}^{n} X_i^k$	$\mu'_k = E(X^k)$

Each μ'_j is a function θ, i.e. $\mu'_j = \mu'_j(\theta_1, \theta_2, \dots, \theta_k)$ for $j = 1, 2, \dots, k$.

Method of Moments Estimators (MME)

We first equate the first k sample moments to the corresponding k population moments,

$$m_1 \approx \mu'_1,$$
$$m_2 \approx \mu'_2,$$
$$\dots$$

$$m_k \approx \mu'_k,$$

Then we solve the resulting systems of simultaneous equations for $\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_k$

Remarks on Method of Moments Estimators

- Very easy to compute
- Always give an estimator to start with
- Generally consistent (Since sample moments are consistent for population moments)
- Not necessarily the best or most efficient estimators

1.2.1.2 Maximum Likelihood Estimators (MLE)

Example

A company receives a certain number of orders per day, and this count appears to follow a Poisson distribution with a parameter λ.

The recorded number of orders over 10 randomly chosen days are: $12, 14, 15, 12, 13, 10, 11, 15, 10, 6$

Deriving the Joint Probability Expression

We can express the probability of observing this particular set of order counts, i.e.,$P(X_1 = 12,\ X_2 = 14, \dots, X_{10} = 6)$ as a function of λ.

$$\begin{aligned}\text{Joint probability of the data} &= P(X_1 = 12,\ X_2 = 14, \dots, X_{10} = 6) \\ &= \frac{e^{-\lambda}\lambda^{12}}{12!}\frac{e^{-\lambda}\lambda^{14}}{14!}\cdots\frac{e^{-\lambda}\lambda^{6}}{6!}\end{aligned}$$

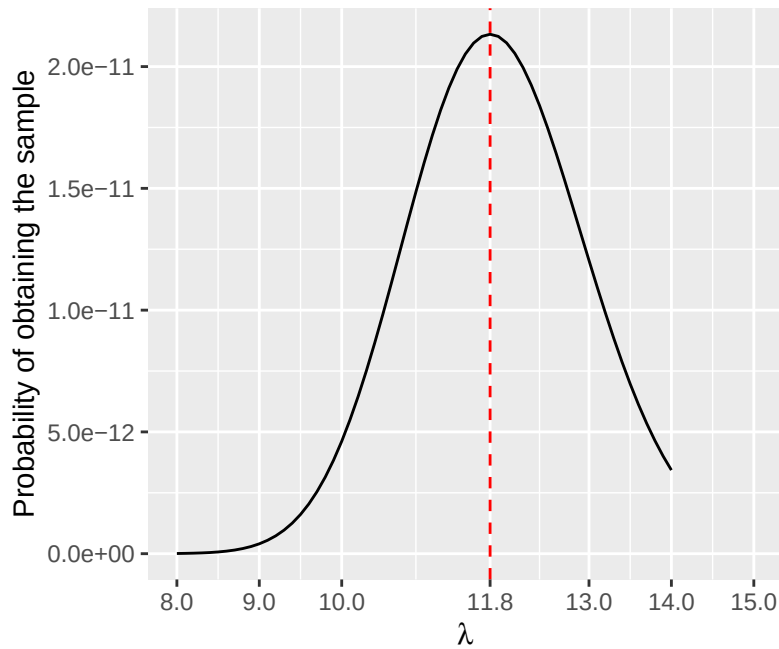


Figure 1.1: Probability of the sample is maximum when $\lambda = 11.8$

- By treating this probability of obtaining the sample as a function of λ, we obtain the **likelihood function**, which represents how likely different values of λ are given the observed data.
- The graphical representation indicates that the likelihood function reaches its maximum when $\lambda = 11.8$.
- Since these data have already occurred, it is very likely that the data have arisen from a Poisson distribution with $\lambda = 11.8$.
- This value is referred to as the **maximum likelihood estimate (MLE)** of λ.

Theory

- In order to define maximum-likelihood estimators, we shall first define the likelihood function.

Definition: Likelihood function

Let $x_1, x_2, \dots, x_n$ be a set of observations of random variables $X_1, X_2, \dots, X_n$ with the joint density of n random variables, say $f_{X_1,X_2,\dots,X_n}(x_1, x_2, \dots, x_n;\ \theta)$. This joint density function, which is considered to be a function of θ is called the **likelihood function of** θ for the set of observations (sample) $x_1, x_2, \dots, x_n$.

In particular, if $x_1, x_2, \dots, x_n$ is a random sample from the density $f(x;\theta)$, then the likelihood function is $f(x_1;\theta)f(x_2;\theta)\dots f(x_n;\theta)$.

Notation

We use the notation $L(\theta;\ x_1, x_2, \dots, x_n)$ for the likelihood function, in order to remind ourselves to think of the likelihood function as a function of θ.

- Likelihood function is seen as a function of θ rather than x
- Likelihood can be viewed as the degree of plausibility.
- An estimate of θ may be obtained by choosing the most plausible value, i.e., where the likelihood function is maximized.

Definition: Maximum Likelihood Estimator

Let $L(\theta) = L(\theta;\ x_1, x_2, \dots, x_n)$ be the likelihood function of θ for the sample $x_1, x_2, \dots, x_n$. Suppose $L(\theta)$ has its maximum when $\theta = \hat{\theta}$.

Then $\hat{\theta}$ is called the **Maximum likelihood estimate of** θ.

The corresponding estimator is called the **Maximum likelihood estimator of** θ.

- Many likelihood functions satisfy regularity conditions; so the maximum likelihood estimator is the solution of the equation

$$\frac{dL(\theta)}{d\theta} = 0$$

Log-likelihood function

Let

$$l(\theta) = ln[L(\theta)].$$

Then, $l(\theta)$ is called the **log-likelihood function**.

- Both $L(\theta)$ and $l(\theta)$ have their maxima at the same value of θ.
- It is sometimes easier to find the maximum of the logarithm of the likelihood and thereby simplify the calculations in finding the maximum likelihood estimate.

Invariance Property of MLE's

If $\hat{\theta}$ is the MLE of θ, then for any function $\tau(\theta)$, the MLE of $\tau(\theta)$ is $\tau(\hat{\theta})$.

1.2.2 Methods of evaluating estimators

- We discussed several methods of obtaining point estimators.
- It is possible that different methods of finding estimators will lead to same estimator or different estimators.
- In this section we discuss certain properties, which an estimator may or may not posses, that will guide us in deciding whether one estimator is better than another.

1.2.2.1 Unbiasedness

Definition: Unbiased estimator

An estimator $\hat{\theta}$ $(= t(X_1, X_2, \dots, X_n))$ is defined to be an **unbiased estimator** of θ if and only if $E(\hat{\theta}) = \theta$

- The difference $E(\hat{\theta}) - \theta$ is called as the bias of $\hat{\theta}$ and denoted by $Bias(\hat{\theta}) = E(\hat{\theta}) - \theta$
- An estimator whose bias is equal to 0 is called **unbiased**.
- This means that, on average, the estimator correctly estimates θ, though it may fluctuate from sample to sample.

Asymptotic unbiasedness

An estimator $\hat{\theta}$ of a parameter θ is said to be *asymptotically unbiased* if its bias

$$Bias(\hat{\theta}) = E(\hat{\theta}) - \theta$$

satisfies

$$\lim_{n \to \infty} Bias(\hat{\theta}) = 0.$$

This means that as the sample size n increases, the bias of the estimator vanishes, making it approximately unbiased for large samples.

1.2.2.2 Consistency

Mean-Squared Error

- The *mean-squared error* is a measure of goodness or closeness of an estimator to the target.

Definition: Mean-squared Error (MSE)

The **mean-squared error** of an estimator $\hat{\theta}$ of θ is defined as $MSE(\hat{\theta}) = E\left[(\hat{\theta} - \theta)^2\right]$

- The MSE measures the average squared difference between $\hat{\theta}$ and θ.
- The MSE is a function of θ and has the interpretation

$$MSE(\hat{\theta}) = Var(\hat{\theta}) + \left[Bias(\hat{\theta})\right]^2$$

- Therefore the MSE incorporates two components, one measuring the variability of the estimator (*precision*) and the other measuring its bias (*accuracy*).
- Small value of MSE implies small combined variance and bias.
- If $\hat{\theta}$ is unbiased, then $MSE(\hat{\theta}) = Var(\hat{\theta})$
- The positive square root of MSE is known as the *root mean squared error* $RMSE(\hat{\theta}) = \sqrt{MSE(\hat{\theta})}$

Consistency

- Estimator $\hat{\theta}$ is said to be consistent for θ if $MSE(\hat{\theta})$ approaches zero as the sample size n approaches ∞.

$$lim_{n\to\infty} E\left[(\hat{\theta} - \theta)^2\right] = 0$$

- Mean-squared error consistency implies that the bias and the variance both approach to zero as n approaches ∞.

1.2.2.3 Best Unbiased Estimator

Why Finding the "Best" Estimator Is Difficult

When estimating an unknown parameter θ, different estimators can be used. The Mean Squared Error (MSE) is a common criterion for comparing estimators because it considers both bias and variance:

$$MSE(\hat{\theta}) = Var(\hat{\theta}) + \left[Bias(\hat{\theta})\right]^2$$

However, a comparison based solely on MSE does not always yield a clear winner. The reason is that:

- There is no single "best" MSE estimator – Some estimators may have lower variance but higher bias, while others have low bias but high variance.
- The class of all possible estimators is too large – Without restrictions, choosing the best estimator is too complex.

Restricting the Class of Estimators

To make this problem tractable, we restrict our focus to only unbiased estimators, which satisfy:

$$E[\hat{\theta}] = \theta$$

By focusing on unbiased estimators, we can now ask: Among all unbiased estimators, which one has the smallest variance? This leads us to the **Uniform Minimum Variance Unbiased Estimator (UMVUE)**.

Uniform Minimum Variance Unbiased Estimator (UMVUE)

Definition

An estimator W^* is a *best unbiased estimator* of $\tau(\theta)$ if it satisfied $E(W^*) = \tau(\theta)$ for all θ and, for any other *estimator* W with $E(W) = \tau(\theta)$, we have $Var(W^*) \leq Var(W)$ for all θ. W^* is also called a *uniform minimum variance unbiased estimator* (UMVUE) of $\tau(\theta)$

In other words, an estimator θ is a *uniform minimum variance unbiased estimator* if

1. It is **unbiased**: $E(\hat{\theta}) = \theta$.
2. It has the **smallest variance** among all unbiased estimators.

Why Do We Care About UMVUE?

Since unbiased estimators fluctuate around the true value, we prefer the one that varies the least. This ensures better precision.

Example

Suppose we want to estimate the mean μ of a population using a sample of size n. Two possible estimators are:

1. **Sample mean:**

$$\hat{\mu}_1 = \frac{1}{n}\sum_{i=1}^{n} X_i$$

which is unbiased because $E(\hat{\mu}_1) = \mu$

2. **Only using the first sample:**

$$\hat{\mu}_2 = X_1$$

which is also unbiased but has **higher variance**.

Although both estimators are unbiased, $\hat{\mu}_1$ is better because it has lower variance.

For a normal distribution $X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2)$, the sample mean:

$$\hat{\mu}_1 = \frac{1}{n}\sum_{i=1}^{n} X_i$$

is the UMVUE for μ, meaning it has the lowest variance among all unbiased estimators.

However, finding the best unbiased estimator (if it exists) is not easy. In some cases, no such estimator exists at all. Even when an unbiased estimator is available, identifying the one with the lowest variance—the best unbiased estimator—requires solving a complex mathematical optimization problem.

1.2.2.3.1 Cramér-Rao Inequality

Why Is This Useful?

Even the best estimator cannot be perfect. The Cramér-Rao bound (CRB) tells us the smallest possible variance for any unbiased estimator. However, an estimator achieving this bound might not exist in certain cases.

Theorem: Cramér-Rao Inequality

Let $X_1, \dots, X_n$ be a sample with pdf $f(x|\theta)$, and let $W(X) = W(X_1, \dots, X_n)$ be any estimator satisfying

$$\frac{d}{d\theta}E[W(X)] = \int_x \frac{\partial}{\partial\theta}[W(x)f(x|\theta)]dx$$

and

$$Var(W(X)) < \infty.$$

Then

$$Var(W(X)) \geq \frac{(\frac{d}{d\theta}E[W(X)])^2}{E\left[\left(\frac{\partial}{\partial\theta}lnf(X|\theta)\right)^2\right]}.$$

In other words, For an unbiased estimator $\hat{\theta}$:

$$Var(\hat{\theta}) \geq \frac{1}{I(\theta)}$$

where $I(\theta)$ is the Fisher Information:

$$I(\theta) = E\left[\left(\frac{\partial}{\partial\theta}lnf(X|\theta)\right)^2\right]$$

The quantity,

$$I(\theta) = E\left[\left(\frac{\partial}{\partial\theta}lnf(X|\theta)\right)^2\right]$$

is called the *information number* or *Fisher information* of the sample as it quantifies the amount of information that the observed data carries about the unknown parameter θ. It gives a bound on the variance of the best unbiased estimator of θ. As the *information number* gets bigger and we have more information about θ, we have a smaller bound on the variance of the best unbiased estimator.

Example

For a normal distribution $X_1, X_2 \dots, X_n \sim N(\mu, \sigma^2)$:, the Fisher Information is:

$$I(\mu) = \frac{n}{\sigma^2}$$

Thus the **Cramér-Rao lower bound** is

$$Var(\hat{\mu}) \geq \frac{\sigma^2}{n}.$$

Since $Var(\hat{\mu}) = \frac{\sigma^2}{n}$, the sample mean $\hat{\mu}$ achieves this bound, making it **the most efficient unbiased estimator.**

1.2.2.4 Sufficiency – Using all available information efficiently

Why do we need sufficiency?

An estimator should make full use of the **available data**. If we can summarize all relevant information into a single function of the data, we can reduce the problem to estimating from that summary.

Definition: Sufficient Statistic

A statistic $T(X)$ is a *sufficient Statistic for a parameter* θ if the conditional distribution of the sample X given the value of $T(X)$ does not depend on θ.

In other words, once you know $T(X)$, **the sample data doesn't give any extra information about** θ.

Factorization Theorem (Easy way to check Sufficiency)

A statistic $T(X)$ is **sufficient** for θ if the likelihood function can be factored as:

$$L(\theta|X) = g(T(X), \theta)h(X)$$

where $g(T(X), \theta)$ depends on θ **only through** $T(X)$

Example

For a normal distribution $X_1, \dots, X_n \sim N(\mu, \sigma^2)$:, the statistic:

$$T = \frac{1}{n} \sum X_i$$

is sufficient for μ.

This means that all the useful information about μ is contained in the sample mean T.

1.2.2.5 Rao-Blackwell Theorem – Improving an estimator

Why Is This Important?

The Rao-Blackwell theorem provides a way to improve an estimator by using a sufficient statistic. This often leads to the UMVUE.

Theorem : Rao-Blackwell Theorem

Let W be any unbiased estimator of $\tau(\theta)$, and let T be a sufficient statistic for θ. Define $\phi(T) = E(W|T)$. then $E(\phi(T)) = \tau(\theta)$ and $Var(\phi(T)) \leq Var(W)$ for all θ; that is , $\phi(T)$ is a uniformly better unbiased estimator of $\tau(\theta)$.

In other words,

Let $\hat{\theta}$ be an unbiased estimator of θ, and let $T(X)$ be a **sufficient statistic**. then the improved estimator:

$$\hat{\theta}^* = E[\hat{\theta}|T(X)]$$

is unbiased and has variance less than or equal to $\hat{\theta}$.

Motivation

If an estimator does not fully use all available information (i.e., it is not based on a sufficient statistic), then it can be improved by conditioning on that statistic.

Example

Suppose we have a normal distribution $X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2)$.

1. The estimator X_1 **alone** is unbiased for μ, but it **ignores most of the data**.
2. The sample mean $\hat{\mu} = \frac{1}{n}\sum(X_i)$ is sufficient for μ.

By the Rao-Blackwell theorem, the improved estimator:

$$E[X_1|\hat{\mu}] = \hat{\mu}$$

has **lower variance** than X_1.

Thus, $\hat{\mu}$ is the UMVUE for μ.

Summary

Concept	Key Idea	Example
Unbiased Estimator	Expected value equals the true parameter.	Sample mean $\hat{\mu}$ is unbiased for μ

Concept	Key Idea	Example
UMVUE	Best unbiased estimator with lowest variance.	Sample mean is UMVUE for μ.
Sufficiency	A statistic that contains all information about θ.	Sample mean is sufficient for μ
Rao-Blackwell Theorem	Improves an estimator using sufficiency.	$\hat{\mu}$ is better than X_1.
Cramér-Rao Inequality	Sets the lower bound for variance.	$Var(\hat{\mu}) \geq \frac{\sigma^2}{n}$.

These ideas are fundamental in statistics and machine learning for choosing the best estimator.

Chapter 1: Tutorial

1. Let $X_1, X_2, \dots, X_n \sim Poisson(\lambda)$. Derive a method of moment estimators for λ.
2. Let $X_1, X_2, \dots, X_n \sim iid\ \ N(\mu, \sigma^2)$, both μ and σ^2 unknown. Derive a method of moment estimators for μ and σ.
3. Let $X_1, X_2, \dots, X_m \sim iid\ \ Bin(n, \theta)$, both n and θ unknown. Derive a method of moment estimators for n and θ.
4. Let $X_1, X_2, \dots, X_n \sim iid\ \ Unif(\theta_1, \theta_2)$, where $\theta_1 < \theta_2$, both unknown. Derive a method of moment estimators for θ_1 and θ_2.
5. Let $X_1, X_2, \dots, X_n \sim iid\ \ Gamma(\alpha, \beta)$, both α and β unknown. Derive a method of moment estimators for α and β.

   The survival time (in weeks) of 20 randomly selected male mouse exposed to 240 units of certain type of radiation are given below.

   $$152, 115, 109, 94, 88, 137, 152, 77, 160, 165, 125, 40, 128, 123, 136, 101, 62, 153, 83, 69$$

   It is believed that the survival times have a gamma distribution. Estimate the corresponding parameters.
6. Let $x_1, x_2, \dots, x_n$ be n random measurements of random variable X with the density function
   $$f_X(x; \lambda) = \lambda x^{\lambda - 1}, \ \ 0 < x < 1, \ \ \lambda > 0$$
   Derive a method of moment estimator for λ.
7. Let $x_1, x_2, \dots, x_n$ be a random sample of size n from a Poisson distribution with parameter λ. Derive the maximum likelihood estimator of λ.
8. Let $x_1, x_2, \dots, x_n$ be a random sample of size n from a normal distribution with mean μ and variance σ^2. Derive the maximum likelihood estimators of μ and σ^2.
9. Let $X_1, X_2, \dots, X_n \sim iid\ Poisson(\lambda)$. Find the MLE of $P(X \leq 1)$
10. Let $X_1, X_2, \dots, X_n \sim iid\ N(\mu, \sigma^2)$. Find the MLE of μ/σ.
11. Let $X_1, X_2, \dots, X_n$ be a random sample from an exponential distribution with the density function
    $$f_X(x; \lambda) = \lambda e^{-\lambda x}, \ \ x > 0.$$
    Is the maximum likelihood estimators of λ unbiased?

Hint

Let $X_1, X_2, \dots, X_n$ be independent and identically distributed (i.i.d.) exponential random variables with rate parameter λ, i.e.,

$$X_i \sim \text{Exp}(\lambda) \quad i = 1, 2, \dots, n.$$

Then, the sum of these n independent exponential random variables

$$S_n = X_1 + X_2 + \cdots + X_n$$

follows a gamma distribution with shape parameter n and rate parameter λ:

$$S_n \sim \text{Gamma}(n, \lambda).$$

Further, for any positive integer $n(> 0)$, Gamma fucntion, $\Gamma(n) = (n-1)!$.

12. Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a normal distribution with mean μ and variance σ^2. Show that $\bar{X}$ and $S^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}$ are unbiased estimators of μ and σ^2, respectively.

13. Let $x_1, x_2, \dots, x_n$ be a random sample of size n from a normal distribution with mean μ and variance σ^2. Consider the maximum likelihood estimators of σ^2. Show the estimator $\hat{\sigma^2} = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n}$ is biased for σ^2, but it has a smaller MSE than $S^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}$.

    Hint

    Distribution of sample variance of a normal distribution

    Suppose that $X_1, X_2, \dots, X_n$ is a random sample from a normal distribution with mean μ and variance σ^2. Let $\bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n}$ be the sample mean and $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$ be the sample variance. Then,

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

14. Let $X_1, X_2, \dots, X_n$ be a random sample from some distribution and $E(X) = \mu$ and $V(X) = \sigma^2$. Show that $\bar{X}$ is a better estimator than X_1 and $\frac{X_1 + X_2}{2}$ for μ in terms of MSE.

15. Let $X_1, X_2, \dots, X_n$ be a random sample from a $N(\mu, \sigma^2)$. Let $\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n}$ and $T = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n}$. Show that $\bar{X}$ is consistent for μ and T is consistent for σ^2.

16. Let $X_1, X_2, \dots, X_n$ be a random sample from a normal distribution

$$X_i \sim N(\mu, \sigma^2)$$

where μ is the unknown mean parameter and σ^2 is a known variance. Consider the following two estimators for μ:

- Estimator 1: The sample mean $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$
- Estimator 2: The first observation X_1.

i. Show that both $\bar{X}$ and X_1 are unbiased estimators of μ.

ii. Compute the variances of $\bar{X}$ and X_1. Which estimator has the smaller variance? Based on this, determine which estimator is the Uniformly Minimum Variance Unbiased Estimator (UMVUE).

iii. Using the Factorization Theorem, determine which of the two estimators is a sufficient statistic for μ.

iv. If X_1 is not sufficient, explain why it can be improved using the Rao-Blackwell Theorem to obtain a better estimator of μ

v. Summarize the findings and explain why the sample mean $\bar{X}$ is the best unbiased estimator of μ.

17. Let $X_1, X_2, \dots, X_n$ be a random sample from a Poisson distribution

$$X_i \sim \text{Poisson}(\lambda), \quad i = 1, 2, \dots, n.$$

Consider the following two estimators for λ:

- Estimator 1: The sample mean $\hat{\lambda_1} = \bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$
- Estimator 2: The mean of the first two observations $\hat{\lambda_2} = \frac{X_1+X_2}{2}$

i. Show that both $\hat{\lambda_1}$ and $\hat{\lambda_2}$ are unbiased estimators of λ.

ii. Compute the variances of both estimators and determine which is more efficient.

iii. Use the Factorization Theorem to determine whether $\bar{X}$ or $\hat{\lambda_2}$ is a sufficient statistic for λ.

iv. Use the Rao-Blackwell Theorem to improve $\hat{\lambda_2}$ as an estimator of λ.

v. Identify the UMVUE (Uniformly Minimum Variance Unbiased Estimator) of λ.

18. A facial recognition algorithm is trained to classify images as "recognized" or "not recognized" based on past training data. It is observed that the probability of correctly recognizing a male face is θ, while the probability of correctly recognizing a female face is θ^2 due to differences in training data representation.

A dataset consisting of n male images and m female images is tested. Among them, x male images and y female images are correctly recognized by the AI model.

i. Derive the log likelihood function for all $n + m$ observations in terms of θ.

ii. Suppose $n = m = 100$, and the AI model correctly recognizes $x = 11$ male faces and $y = 2$ female faces. Derive the Maximum Likelihood Estimate (MLE) of θ.

iii. To address bias in the facial recognition model, a calibration function is introduced: $\mu = g(\theta) = 200 - 20Z_\theta$. Find the MLE for μ

2 Interval Estimation

- Under point estimation of a parameter θ, the inference is a guess of a **single value** as the value of θ.
- Instead of making the inference of estimating the true value of the parameter to be a point, under interval estimation we make the inference of estimating that the true value of the parameter is contained in **some interval.**

2.1 What is gained by using an Interval Estimator?

Example

- For a sample X_1, X_2, X_3, X_4 from a $N(\mu, 1)$, an interval estimator of μ is $[\bar{X}-1, \bar{X}+1]$.
- This means that we will assert that μ is in this interval.
- In the previous section (Point estimation) we estimated μ with $\bar{X}$.
- But now we have the less precise estimator $[\bar{X}-1, \bar{X}+1]$.
- Under interval estimation, by giving up some precision in our estimate (or assertion about μ), we try to gain some confidence , or assurance that our assertion is correct.

Explanation

- When we estimate μ by $\bar{X}$, the probability that the estimator exactly equaled the value of the parameter being estimated is zero (Why? the probability that a continuous random variable equals any value is 0), *i.e.* $P(\bar{X} = \mu) = 0$.
- However, with an interval estimator, we have a positive probability of being correct.
- The probability that μ is covered by the interval $[\bar{X}-1, \bar{X}+1]$ can be calculated as

$$P(\mu \in [\bar{X}-1, \bar{X}+1]) = P(\bar{X}-1 \leq \mu \leq \bar{X}+1)$$

$$= P(-1 \leq \bar{X}-\mu \leq 1)$$

$$= P(-2 \leq \frac{\bar{X}-\mu}{\sqrt{1/4}} \leq 2)$$

$$= P(-2 \leq Z \leq 2) \quad \left(\frac{\bar{X} - \mu}{\sqrt{1/4}} \text{ is standard normal} \right)$$
$$= 0.9544.$$

- Therefore now we have over 95% chance of covering the unknown parameter with the interval estimator.
- By moving for a point to an interval we have scarified some precision in our estimate. But it has resulted in increased confidence that our assertion is correct.
- The purpose of using an interval estimator rather than a point estimator is to have some guarantee of capturing the parameter of interest.

2.2 Definition of confidence interval

Definition

Let $X_1, X_2, \dots, X_n$ be a random sample from a distribution with parameter θ. Let $T_1 = g(X_1, X_2, \dots, X_n)$, and $T_2 = h(X_1, X_2, \dots, X_n)$ be two statistics satisfying $T_1 \leq T_2$ for which $P(T_1 < \theta < T_2) = \gamma$, where γ does not depend on θ. Then, the random interval (T_1, T_2) is called a 100γ **percent confidence interval for** θ; γ is called the confidence coefficient; and T_1 T_2 are called the lower and upper confidence limits, respectively, for θ.

Suppose that $x_1, x_2, \dots, x_n$ is a realization of $X_1, X_2, \dots, X_n$ and let $t_1 = g(x_1, x_2, \dots, x_n)$, and $t_2 = h(x_1, x_2, \dots, x_n)$. Then the *numerical* interval (t_1, t_2) is also called a 100γ **percent confidence interval for** θ.

2.3 Interpretation of confidence intervals

- Consider the probability statement $P(\bar{X} - 1.18 \leq \mu \leq \bar{X} + 1.18) = 0.95$.
- The above probability statement implies that the random interval $(\bar{X} - 1.18, \bar{X} + 1.18)$ includes the unknown true mean μ with probability 0.95.

2.4 Methods of finding interval estimators

2.4.1 Pivotal Quantity Method

Definition: Pivotal Quantity

Let $X_1, X_2, \dots, X_n$ be a random sample from the density $f(.;\theta)$. Let $Q = q(X_1, X_2, \dots, X_n; \theta)$; that is, let Q be a function of $X_1, X_2, \dots, X_n$ and θ. If Q has a distribution that does not depend on θ, then Q is defined to be a *pivotal quantity*

Pivotal Quantity method

If $Q = q(X_1, X_2, \dots, X_n; \theta)$ is a pivotal quantity and has a probability density function, then for any fixed $0 < \gamma < 1$ there will exist q_1 and q_2 depending on γ such that $P[q_1 < Q < q_2] = \gamma$. Now, if for each possible sample value $(x_1, x_2, \dots, x_n)$, $q_1 < q(x_1, x_2, \dots, x_n; \theta) < q_2$ if and only if $t_1(x_1, x_2, \dots, x_n) < \tau(\theta) < t_2(x_1, x_2, \dots, x_n)$ for functions t_1 and t_2 (not depending on θ), then (T_1, T_2) is a 100γ percent confidence interval for $\tau(\theta)$, where $T_1 = t_1(X_1, X_2, \dots, X_n)$ and $T_2 = t_2(X_1, X_2, \dots, X_n)$.

2.5 Methods of evaluating interval estimators

- Coverage probability
- Size (expected length)

Chapter 2: Tutorial

1. Let $X_1, X_2, \ldots, X_2 5$ be a random sample from a $N(\mu, 9)$. Find a 95% confidence interval for μ.

2. Let $X_1, X_2, \ldots, X_n$ be a random sample from a $N(\mu, \sigma^2)$, both μ and σ are unknown. Construct a $100(1-\alpha)\%$ $(0 < \alpha < 1)$ confidence interval for μ.

3. Let $X_1, X_2, \ldots, X_n$ be a random sample from a $N(\mu, \sigma^2)$,. Construct a $100(1-\alpha)\%$ $(0 < \alpha < 1)$ confidence interval for σ^2.

4. A random sample of 144 with a mean of 100 and standard deviation of 60 is taken from a population of 1000. Construct a 95% confidence interval for the unknown population mean.

5. A manager wishes to estimate the mean number of minutes that workers take to complete a particular manufacturing process with ± 3 min and with 90% confidence. From past experience, the manager knows that the standard deviation σ is 15 min. What is the minimum required sample size $(n > 30)$ for this estimation.

6. A state Education department finds that in a random sample of 100 persons who attended college, 40 received a college degree. find the 99% confidence interval for the proportion of college graduates out of all the persons who attended college.

7. A random sample of 25 with a mean of 80 and a standard deviation of 30 is taken from a population of 1000 that is normally distributed.

a) Find 90% confidence interval for the unknown population mean.

b) Find 95% confidence interval for the unknown population mean.

c) Find 99% confidence interval for the unknown population mean.

d) what does the difference in the results to parts a), b), and c) indicate?

Cumulative Standard Normal Distribution Table

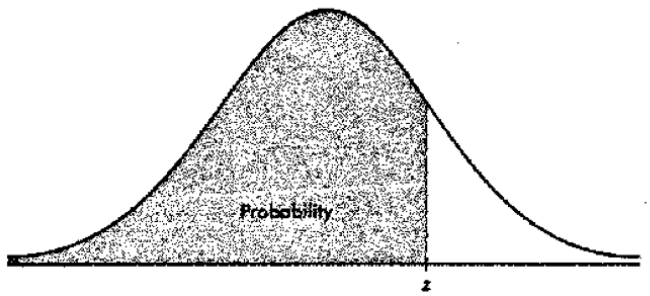


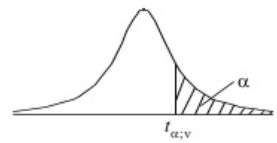
TABLE A: STANDARD NORMAL PROBABILITIES (CONTINUED)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

Student's t distribution Table

Table of the Student's *t*-distribution

The table gives the values of $t_{\alpha;\nu}$ where
$\Pr(T_\nu > t_{\alpha;\nu}) = \alpha$, with ν degrees of freedom



ν \ α	0.1	0.05	0.025	0.01	0.005	0.001	0.0005
1	3.078	6.314	12.076	31.821	63.657	318.310	636.620
2	1.886	2.920	4.303	6.965	9.925	22.326	31.598
3	1.638	2.353	3.182	4.541	5.841	10.213	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	1.319	1.714	2.069	2.500	2.807	3.485	3.767
24	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	1.296	1.671	2.000	2.390	2.660	3.232	3.460
120	1.289	1.658	1.980	2.358	2.617	3.160	3.373
∞	1.282	1.645	1.960	2.326	2.576	3.090	3.291

Chi-Square Distribution Table

Degrees of freedom	Area in upper tail									
	.995	.99	.975	.95	.9	.1	.05	.025	.01	.005
1	0.0000393	0.0001571	0.0009821	0.0039322	0.0157907	2.7055	3.8415	5.0239	6.6349	7.8794
2	0.010025	0.020100	0.050636	0.102586	0.210721	4.6052	5.9915	7.3778	9.2104	10.5965
3	0.07172	0.11483	0.21579	0.35185	0.58438	6.2514	7.8147	9.3484	11.3449	12.8381
4	0.20698	0.29711	0.48442	0.71072	1.06362	7.7794	9.4877	11.1433	13.2767	14.8602
5	0.41175	0.55430	0.83121	1.14548	1.61031	9.2363	11.0705	12.8325	15.0863	16.7496
6	0.67573	0.87208	1.23734	1.63538	2.20413	10.6446	12.5916	14.4494	16.8119	18.5475
7	0.98925	1.23903	1.68986	2.16735	2.83311	12.0170	14.0671	16.0128	18.4753	20.2777
8	1.34440	1.64651	2.17972	2.73263	3.48954	13.3616	15.5073	17.5345	20.0902	21.9549
9	1.73491	2.08789	2.70039	3.32512	4.16816	14.6837	16.9190	19.0228	21.6660	23.5893
10	2.15585	2.55820	3.24696	3.94030	4.86518	15.9872	18.3070	20.4832	23.2093	25.1881
11	2.60320	3.05350	3.81574	4.57481	5.57779	17.2750	19.6752	21.9200	24.7250	26.7569
12	3.07379	3.57055	4.40378	5.22603	6.30380	18.5493	21.0261	23.3367	26.2170	28.2997
13	3.56504	4.10690	5.00874	5.89186	7.04150	19.8119	22.3620	24.7356	27.6882	29.8193
14	4.07466	4.66042	5.62872	6.57063	7.78954	21.0641	23.6848	26.1189	29.1412	31.3194
15	4.60087	5.22936	6.26212	7.26093	8.54675	22.3071	24.9958	27.4884	30.5780	32.8015
16	5.14216	5.81220	6.90766	7.96164	9.31224	23.5418	26.2962	28.8453	31.9999	34.2671
17	5.69727	6.40774	7.56418	8.67175	10.08518	24.7690	27.5871	30.1910	33.4087	35.7184
18	6.26477	7.01490	8.23074	9.39045	10.86494	25.9894	28.8693	31.5264	34.8052	37.1564
19	6.84392	7.63270	8.90651	10.11701	11.65091	27.2036	30.1435	32.8523	36.1908	38.5821
20	7.43381	8.26037	9.59077	10.85080	12.44260	28.4120	31.4104	34.1696	37.5663	39.9969
21	8.03360	8.89717	10.28291	11.59132	13.23960	29.6151	32.6706	35.4789	38.9322	41.4009
22	8.64268	9.54249	10.98233	12.33801	14.04149	30.8133	33.9245	36.7807	40.2894	42.7957
23	9.26038	10.19569	11.68853	13.09051	14.84795	32.0069	35.1725	38.0756	41.6383	44.1814
24	9.88620	10.85635	12.40115	13.84842	15.65868	33.1962	36.4150	39.3641	42.9798	45.5584
25	10.51965	11.52395	13.11971	14.61140	16.47341	34.3816	37.6525	40.6465	44.3140	46.9280
26	11.16022	12.19818	13.84388	15.37916	17.29188	35.5632	38.8851	41.9231	45.6416	48.2898
27	11.80765	12.87847	14.57337	16.15139	18.11389	36.7412	40.1133	43.1945	46.9628	49.6450
28	12.46128	13.56467	15.30785	16.92788	18.93924	37.9159	41.3372	44.4608	48.2782	50.9936
29	13.12107	14.25641	16.04705	17.70838	19.76774	39.0875	42.5569	45.7223	49.5878	52.3355
30	13.78668	14.95346	16.79076	18.49267	20.59924	40.2560	43.7730	46.9792	50.8922	53.6719
40	20.70658	22.16420	24.43306	26.50930	29.05052	51.8050	55.7585	59.3417	63.6908	66.7660
50	27.99082	29.70673	32.35738	34.76424	37.68864	63.1671	67.5048	71.4202	76.1538	79.4898
60	35.53440	37.48480	40.48171	43.18797	46.45888	74.3970	79.0820	83.2977	88.3794	91.9518
70	43.27531	45.44170	48.75754	51.73926	55.32894	85.5270	90.5313	95.0231	100.4251	104.2148
80	51.17193	53.53998	57.15315	60.39146	64.27784	96.5782	101.8795	106.6285	112.3288	116.3209
90	59.19633	61.75402	65.64659	69.12602	73.29108	107.5650	113.1452	118.1359	124.1162	128.2987
100	67.32753	70.06500	74.22188	77.92944	82.35813	118.4980	124.3421	129.5613	135.8069	140.1697

2.6 Summary

Condition	Sample size	Target Parameter	Test Statistic
Population is Normal, σ known	Large or Small	μ	$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$
Population is Normal, σ unknown	Small	μ	$\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t_{n-1}$
Population is Normal, σ unknown	Large	μ	$\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t_{n-1}$ or $\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim N(0,1)$
Population is Unknown, σ known	Sample size is large ($n \geq 30$)	μ	Central Limit Theorem $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$
Population is Unknown, σ unknown	Sample size is large ($n \geq 30$)	μ	$\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t_{n-1}$ or $\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim N(0,1)$ (Since n is large there is a little difference between these values)
Population is Binomial	Sample size is large $np \geq 5$ and $n(1-p) \geq 5$	θ	$\frac{p-\theta}{\sqrt{\frac{p(1-p)}{n}}} \sim N(0,1)$
Population is Normal	Large or Small	σ^2	$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$

3 Hypothesis Testing

- Hypothesis testing about population characteristics is another fundamental aspect of statistical inference.

  Definition: Hypothesis

  A *hypothesis* is a statement about a **population parameter**.

- In testing hypothesis, we start by making *an assumption* with regard to an unknown population characteristic.
- We then take a random sample from the population, and on the basis of the corresponding sample characteristic, we either accept or reject the hypothesis with a particular degree of confidence.

3.1 Null and alternative hypotheses

- Statistical hypotheses consist of the **null hypothesis** and the **alternative hypothesis,** which together contain all the possible outcomes of the experiment or study.

  null hypothesis

- Generally, the *null hypothesis* states that a treatment has *no effect* (and has led to the term "null" hypothesis).

  alternative hypothesis

- Usually, alternative hypothesis is the negation, or complement, of the null hypothesis.
- Usually it is the hypothesis that the researcher is interested in proving.
- The null hypothesis is denoted H_0, and the alternative is denoted H_1.
- Let θ is a population parameter.
- the general format of the null hypothesis and alternative hypothesis is

  $$H_0 : \theta \in \Theta_o \text{ and } H_1 : \theta \in \Theta_o^c$$

  where Θ_o is some subset of the parameter space and Θ_o^c is its complement.

3.2 Different types of Hypotheses

(1) **simple hypotheses:** both H_0 and H_1 consist of only one probability distibution

$$H_0 : \theta = \theta_0 \text{ vs } H_1 : \theta = \theta_1$$

(2) **composite hypothesis:** either H_0 or H_1 has more than one probability distribution

- *one sided*: $H_0 : \theta \geq \theta_0$ vs $H_1 : \theta < \theta_0$
- *one sided*: $H_0 : \theta \leq \theta_0$ vs $H_1 : \theta > \theta_0$
- *two sided*: $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$

3.2.1 Writing H_0 and H_1

1. The equality sign (eq: $=, \leq, or \geq$)) always goes to H_0
2. $H_0 :$ $H_1 :$ which is to be tested (claim to be tested)
3. $H_0 :$ initially favoured one before collecting information $H_1 :$

3.2.2 Two tailed test and one tailed test

Two tailed test

- Two tailed tests always use $=$ and $\neq$ in the statistical hypotheses.
- Alternative hypothesis allows for either the $>$ or $<$ possibility.
- Here the research is interested in testing deviations from the null in two directions.

One tailed test

- One tailed tests are always directional, and the alternative hypothesis uses either the $>$ or $<$ sign.

3.3 Rejection region

A critical region, also known as the rejection region, is a set of values for the test statistic for which the null hypothesis is rejected.

Figure 3.1: Rejection and nonrejection regions

3.4 Errors in testing hypotheses-type I and type II error

- Samples are used to determine whether to reject H_0 or not.
- Since the decision to reject H_0 or not, is based on incomplete (i.e sample information), there is always a possibility of making an incorrect decision.
- We can make two types errors in testing a hypothesis.
  - *Type I error:* if H_0 is true, but the test incorrectly reject H_0. (reject H_0/ H_0 true)
  - *Type II error:* if H_0 is false, but the test incorrectly accept H_0. (accept H_0/ H_1 true)

	Decision	
Truth	Accept H_0	Reject H_0
H_0	Correct decision	**Type I error** (α)
H_1	**Type II error** (β)	Correct decision (*power*)

- Typically, a **Type I error is the most serious error** and that is what we try to control.

3.4.1 Significance level

- The probability of committing a type I error is called α or *significance level* (level of significance)
- α equals the area under the curve that is in the rejection region beyond the critical value(s).
- The value of α is always set before the experiment or study is undertaken.
- $1 - \alpha$ is called the *level of confidence* of the test.
- Common values of α are 0.05, 0.01, 0.10 and 0.001.

3.4.2 Power of a test

- The probability of committing a Type II error is β.
- **Power** of a test is equal to $1 - \beta$.
- *i.e.*, the probability of a test rejecting the null hypothesis when the null hypothesis is false.

3.5 Testing hypotheses

- Typically, the hypothesis testing process is presented in terms of an eight-step approach.

  Task 1: Hypothesise

  Step 1: Establish null and alternative hypotheses.

  Task II: Test

  Step 2: Determine the appropriate statistical test

  Step 3: Set the value of α, the Type I error rate.

  Step 4: Establish the decision rule

  Step 5: Gather sample data

  Step 6: Analyse the data

  Task III: Take statistical action

  Step 7: Reach a statistical conclusion.

  Task IV: Determine the business implications

  Step 8: Make a business decision.

- This process of testing hypothesis is referred as the HTAB system, where HTAB is an abbreviation for Hypothesise, Test, Action, Business.

3.6 Methods of testing hypotheses

3.6.1 Using the critical value method to test hypotheses

- A method of testing hypotheses by comparing the sample statistic with the critical value in order to reach a conclusion about rejecting or failing to reject the null hypothesis.
- Critical region: Set of all values of the test statistic that would cause us to reject null hypothesis.

3.6.2 Using the p-value to test hypotheses

- Get the decision by comparing α value with p-value.
- **p-value:** Probability of observing a *sample statistic* at least as extreme as the observed test statistic computed *under the assumption that* H_0 is true.
- The p-value defines the smallest value of α for which the null hypothesis can be rejected

Example

- If the p-value of a test is 0.038, the null hypothesis cannot be rejected at $\alpha = 0.01$ because 0.038 is the smallest value of alpha for which the null hypothesis can be rejected. However, the null hypothesis can be rejected for $\alpha = 0.05$.
- If the p-value of a test is 0.0207, , the researcher would reject the null hypothesis for $\alpha = 0.05$ or $\alpha = 0.10$ or any value more than 0.0207.
- **Virtually every statistical computer program yields this probability (p-value)**

3.6.3 Using confidence intervals to test hypotheses

- Confidence intervals also give some indication about the decision.

3.7 Hypothesis Testing for a Single Population

Scenario	Target Parameter	Conditions	Test Statistic	Test Name
Test for population mean (σ known)	μ	Population normal (any size) OR unknown with large n ($n \geq 30$)	$z = \frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$	One-sample z-test
Test for population mean (σ unknown, small n)	μ	Population normal, small sample ($n < 30$)	$t = \frac{\bar{x}-\mu_0}{S/\sqrt{n}} \sim t_{n-1}$	One-sample t-test
Test for population mean (σ unknown, large n)	μ	Population unknown/normal, large sample ($n \geq 30$)	$t = \frac{\bar{x}-\mu_0}{S/\sqrt{n}} \sim t_{n-1}$ or $N(0,1)$	One-sample t-test (approx. z-test)
Test for population proportion	θ	Binomial population, $n\theta_0 \geq 5$ and $n(1-\theta_0) \geq 5$	$z = \frac{\hat{p}-\theta_0}{\sqrt{\theta_0(1-\theta_0)/n}} \sim N(0,1)$	One-sample z-test for proportion
Test for population variance	σ^2	Population normally distributed	$\chi^2 = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$	Chi-square test for variance

3.8 Application of Hypothesis Testing: Choosing the Right Statistical Test

			Dependent Variable (Y)			
			Categorical		Continuous	
			1 Variable, 2 Categories	1 Variable, >2 Categories	1 Variable	>1 Variable
Independent Variable/ Predictor/ Covariate (X)	Categorical	1 Variable, 2 Categories, Between-subjects	Chi-square Test		Independent t Test	
		1 Variable, 2 Categories, Within-subjects			Paired t Test	
		1 Variable, >2 Categories, Between-subjects			One-Way ANOVA	One-Way MANOVA
		1 Variable, >2 Categories, Within-subjects			Repeated Measures ANOVA	Repeated Measures MANOVA
		>1 Variable, All Categorical, Between Subjects	Binomial Logistic Regression with Categorical Predictors	Multinomial Logistic Regression	Factorial ANOVA	Factorial MANOVA
		>1 Variable, All Categorical, Mixed Within- & Between- Subjects			Mixed-Design ANOVA	Mixed-Design MANOVA
		>1 Variable, Mixed Categorical & Continuous			One-Way ANCOVA	One-Way MANCOVA
	Continuous	1 Variable	Binomial Logistic Regression		Simple Linear Regression	Multivariate Linear Regression
		>1 Variable			Multiple Linear Regression	

Figure 3.2: Selection of statistical tests in hypothesis testing based on variable type

- The **dependent variable** (response variable) is the variable that is being measured or tested in an experiment.
- The **independent variable** (explanatory) is the variable the experimenter manipulates or changes, and is assumed to have a direct effect on the **dependent variable**.
- **Between-subjects** (or between-groups) study design: different people test each condition, so that each person is only exposed to one condition.
- **Within-subjects** (or repeated-measures) study design: the same person tests all the conditions.

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Chapter 3: Tutorial

1. A recent industry report from over a decade ago found that the average annual income of freelance software developers was \$74,914. An IT economics researcher believes that this figure may have changed over time due to shifts in the tech job market, the rise of remote work, and platform-based freelancing. To investigate this, the researcher collects a random sample of 112 freelance software developers and records their annual incomes. Assume that the population standard deviation of annual income for freelance developers is \$14,530. The sample of 112 developers yields a mean annual income of \$78,695. The researcher wants to check whether the average income of freelance software developers has significantly changed from the previously reported value of \$74,914. Use a significance level of $\alpha = 0.05$ to test the researcher's belief.

2. An IT company manages a server system that is expected to maintain an average response time of 250 milliseconds per request. A system administrator suspects that the server may no longer be maintaining this expected performance. To investigate, a random sample of 20 recorded response times from the previous day was selected. The sample yielded a mean response time of 255.1 milliseconds and a standard deviation of 21.93 milliseconds. Assuming that the response times are normally distributed, test whether the server is operating outside the expected performance level. Use a significance level of $\alpha = 0.05$.

3. In professional photography, especially under low-light conditions, image noise is a common challenge. Traditionally, standard noise reduction software resulted in an average image clarity score (on a 0–100 scale) of 471, based on expert assessments after post-processing. Recently, an AI-powered noise reduction tool has been introduced, claiming to improve image clarity beyond the standard level. To evaluate this claim, a photography lab applied the new AI tool to a random sample of 23 photographs taken in challenging lighting conditions. The resulting clarity scores had a sample mean of 498.78 and a sample standard deviation of 46.94. Assuming the clarity scores are normally distributed, test at the 5% level of significance whether the AI tool provides a significant improvement in image clarity over the traditional method

4. An AI company claims that only 8% of images processed through their noise reduction algorithm still show visible noise artifacts. A quality assurance researcher wants to test this claim. To evaluate the system, the researcher randomly selects a sample of 200 processed images and inspects them for noticeable noise. It is found that 33 images still contain visible noise artifacts. Using a significance level of 0.10, test whether the proportion of flawed images is significantly different from the claimed 8%.

5. A university survey reveals that 17% of undergraduates use generative AI tools (such as ChatGPT) during online exams, often doing so without a proper understanding of the course material or taking the time to critically assess the responses provided. This

overreliance not only undermines genuine learning but also reflects a lack of academic discipline — a habit that can severely affect one's long-term development and ability to compete in the professional world. A researcher believes this issue is more prevalent among undergraduates in a particular degree programme at a university. To test this, a random sample of 550 students from that programme was surveyed, with 115 reporting over-reliance on AI during their most recent online exam. Using a significance level of 0.05, test if the proportion of students in this programme is higher than the university average of 17%.

6. An aviation maintenance company has been working diligently to implement a just-in-time (JIT) inventory system for its aircraft component repair line. The final product requires the installation of specific engine components at a particular workstation during the repair process. With the JIT system, the company aims to minimize the number of components waiting at the station to be installed, as storage costs for excess inventory are very high. Ideally, the components would arrive just as the technician needs them. However, due to supplier and transportation variables, there is often some buildup of component inventory. The company expects that, on average, about 20 engine components will be at the station. However, the maintenance manager does not want the variance of this inventory to exceed 4. On a given day, the number of components at the workstation is recorded at eight different times, and the sample variance is calculated as 20.9821. Assume the number of components is normally distributed. Use a significance level of 0.05 to test whether the variance exceeds the acceptable limit.

7. A tech company specializing in AI solutions uses automated project management tools to assign tasks and track project timelines for its 37 employees. Despite these tools, unexpected client requests and urgent model updates often result in overtime work. Historically, the company estimates around 50 total overtime hours per week, with a variance of 25 *hours*. To assess whether the variability in weekly overtime has shifted from this historical estimate, management analyzes system-logged overtime data over 16 randomly selected weeks. The sample variance is 28.1. Assuming weekly overtime hours are normally distributed, use a significance level of 0.10 to test whether there is evidence that the variance is no longer 25.

4 Analysis of Variance and Regression

- The methods of Analysis of Variance (ANOVA) and regression analysis are fundamental tools in statistical modeling, enabling researchers to explore relationships between variables and test hypotheses about group differences and linear associations.
- This chapter introduces the principles and applications of ANOVA, which is used to determine whether the means of multiple groups differ significantly, particularly when dealing with categorical explanatory variables.
- Building on this foundation, the chapter then explores Simple Linear Regression, which models the relationship between a single predictor and a response variable, and extends to Multiple Linear Regression, where two or more predictors are used to explain or predict the behavior of a dependent variable.
- Together, these techniques form a powerful framework for data analysis, supporting both inference and prediction in a wide range of scientific and practical contexts.

4.1 One-Way Analysis of Variance (ANOVA) and multiple comparisons

- The One-Way Analysis of Variance (ANOVA) is a parametric statistical method used to test whether there are statistically significant differences between the means of three or more independent groups.
- It extends the logic of the independent samples t-test, which is limited to comparing the means of two groups, by simultaneously analyzing variability among multiple groups within a single unified framework.
- An **experimental design** is a plan and structure used to test hypotheses, where the researcher controls or manipulates one or more variables. It includes both independent and dependent variables..
- Many different types of experimental designs are available to researchers. In this chapter, we focus on one of the simplest experimental designs: the completely randomized design. Other design types such as randomized block designs and factorial experiments are beyond the scope of this chapter.

- In an experimental design, an **independent variable** may either be a **treatment variable** or a **classification variable**.
- A **treatment variable** is one that the experimenter controls or modifies during the experiment.
- A **classification variable** refers to a characteristic of the experimental subjects that existed prior to the experiment and is not a result of the experimenter's manipulation or control.
- Independent variables are also referred to as **factors**.
- **Levels** are the subcategories or specific values of the independent variable used by the researcher in the experimental design.
- The other key variable in an experimental design is the **dependent variable**.
- A **dependent variable** is the observed response to the different levels of the independent variable.
- A completely randomized design includes only one independent variable, which has two or more treatment levels or categories.
- When the independent variable has only two treatment levels or categories, the design is equivalent to that used for comparing the means of two independent populations, where the independent t-test is typically used for analysis.
- A Completely randomized design is analysed by a **one-way analysis of variance (ANOVA)**.
- In general, when analyzing k samples using one-way ANOVA, the hypotheses tested are:

  H_0: $\mu_1 = \mu_2 = \mu_3 = \cdots = \mu_k$

  H_1: At leats one of the means is different from the others
- In one-way ANOVA, the total variance in the data is partitioned into the following two components:

  1. the variance resulting from the treatment
  2. the error variance or the portion fo the total variance unexplained by the treatment
- As part of the analysis, the total sum of squares (i.e., the total deviation of observations from the overall mean) is divided into two additive and independent parts: the sum of squares due to treatment (tretment variation) and the sum of squares due to error (error variation).

$$\sum_{i=1}^{n_j}\sum_{j=1}^{C}(x_{ij}-\bar{x})^2 = \sum_{j=1}^{C} n_j(\bar{x_j}-\bar{x})^2 + \sum_{i=1}^{n_j}\sum_{j=1}^{C}(x_{ij}-\bar{x_j})^2$$

where:

i = particular member of a treatment level

j = a treatment level

C = number of treatment levels

n_j = number of observations in a given treatment level

$\bar{x}$ = grand mean

$\bar{x_j}$ = mean of the treatment group or level

x_{ij} = individual value

Contd….

- ANOVA compares the relative sizes of the treatment variation and the error variation.
- If a significant difference in treatments is present, the treatment variation should be large relative to the error variation.
- Several important assumptions underlie analysis of variance.

  Assumption #1: Observations are drawn from normally distributed populations.

  That is each sample is taken from a normally distributed population.

  Assumption #2: Variances of the populations are equal.

  That is the variance of data in the different groups should be the same.

  Assumption #3: Observations represent random samples from the populations.

4.1.1 Formulas for computing a one-way ANOVA

$$SST = \sum_{i=1}^{n_j} \sum_{j=1}^{C} (x_{ij} - \bar{x})^2$$

$$SSC = \sum_{j=1}^{C} n_j (\bar{x_j} - \bar{x})^2$$

$$SSE = \sum_{i=1}^{n_j} \sum_{j=1}^{C} (x_{ij} - \bar{x_j})^2$$

- In the context of ANOVA, degrees of freedom (df) refer to the number of independent pieces of information available to estimate variance components and to test hypotheses.

$$df_T = N - 1$$

$$df_C = C - 1$$

$$df_E = N - C$$

- Mean square is an average and is computed by dividing the sum of squares by the degrees of freedom.

$$MSC = \frac{SSC}{df_C}$$

$$MSE = \frac{SSE}{df_E}$$

- The **F value (ANOVA)** is the ratio of the treatment variance (MSC) to the error variance (MSE).

$$F = \frac{MSC}{MSE}$$

Example

Automated hand sanitization plays a crucial role in promoting hygiene in public and private spaces.. In response to this, a company designed a portable hand sanitizer dispenser robot. Suppose the quality control engineer of the company wants to analyse the response time of four portable hand sanitizer dispenser robots before releasing them to the market.

1. What is the independent variable in this design?

   Portable hand sanitizer dispenser robots

2. What is the dependent variable in this design?

   Response time

3. Is there a significant difference in the mean response time of 24 tasks carried out by the four portable hand sanitizer dispenser robots?

Robot 1	Robot 2	Robot 3	Robot 4
6.33	6.26	6.44	6.29
6.26	6.36	6.38	6.23
6.31	6.23	6.58	6.19
6.29	6.27	6.54	6.21
6.4	6.19	6.56	
	6.50	6.34	
	6.19	6.58	
	6.22		

Refer Python Lab session

4.2 Regression analysis

- **Regression analysis** is a statistical technique for investigating and **modeling the relationship between variables**.
- There are numerous applications of regression in many fields.

Examples

- A production company may need to determine how its sales related to advertising
- How the growth of the bacteria is related to moisture level of the environment.
- The relationship between blood pressure and the age of a person.
- the relationship between transaction time and transaction amount in fraud detection.

- Usually, the first step in regression analysis is to construct a scatter plot (or scatter matrix).
- Graphing the data in a scatter plot yields preliminary information about the *shape* and *spread* of the data.

4.2.1 Simple Linear Regression

- The most elementary regression model is called **simple linear regression**.
- Is is also known as *bivariate linear regression*, which means that it involves only two variables.
- One variable is predicted by another variable.
- The variable to be predicted is called the *dependent variable* an is denoted by y.
- The *predictor* is called the *independent variable* or *explanatory variable* and is denoted by x
- In simple *linear* regression analysis, only a strait-line relationship between two variables is examined.
- Nonlinear relationships and regression models with more than one independent variable can be explored by using multiple regression models.

4.2.1.1 Determining the equation of the regression line

- The first step in determining the equation of the regression line that passes through the sample data is to establish the equation's form.

- In mathematics, the equation of a line can be written as

$$y = mx + c$$

where: m = slope of the line
$c = y$ intercept of the line.

- In statistics, the slope-intercept form of the equation of the regression line through the population points is:

$$\hat{y} = \beta_0 + \beta_1 x$$

where: $\hat{y}$ = the predicted value of y
β_0 = the population y intercept
β_1 = the population slope.

- For any specific dependent variable value, y_i:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

where: x_i = the value of the independent variable for the ith value
y_i = the value of the dependent variable for the ith value
β_0 = the population y intercept
β_1 = the population slope
ϵ_i = the error of prediction for the ith value.

- Unless the points being fitted by the regression equation are in perfect alignment, the regression line will miss at least some of the points.

- In the above equation, ϵ_i represents the error of the regression line in fitting these points. If a point is on the regression line, $\epsilon_i = 0$

4.2.1.2 Deterministic models vs probabilistic models

- These mathematical models can be either deterministic models or probabilistic models.

Deterministic models

- Deterministic models are *mathematical models that produces an* **exact** *output for a given input*
- For example, consider the equation of a regression line is:

$$\hat{y} = 1.68 + 2.40x$$

.

For a value of $x = 5$, the exact predicted value of y is

$$\hat{y} = 1.68 + 2.40 \times 5 = 13.68$$

.

- However, the most of the time the values of y will not equal exactly the values yields by the equations.

- Random error will occur in the prediction of the y values for values of x, because it is likely that the variable x does not explain all the variability of the variable y.

Example

- Suppose we want to predict the sales volume (y) for a mobile phone company through regression analysis by using the annual amount of advertising (in Rupees) (x) as the predictor.
- Although sales are often related to advertising, there can be other factors related to sales that are not accounted for by the amount of advertising.
- Therefore, a regression model to predict sales volume by the amount of advertising probably involves some error.
- For this reason, in regression, we present the general model as a probabilistic model.

Probabilistic models

- A probabilistic model is *one that includes an error term that allows for the y values to vary for any given value of x.*
- The deterministic regression model is

$$y = \beta_0 + \beta_1 x$$

- The probabilistic regression model is

$$y = \beta_0 + \beta_1 x + \epsilon.$$

- $\beta_0 + \beta_1 x$ is the deterministic portion of the probabilistic model, $\beta_0 + \beta_1 x + \epsilon$.
- In deterministic model, all points are assumed to be on the line and in all cases ϵ is zero.

4.2.1.3 Least squares estimation of the parameters

- The parameters β_0 and β_1 are unknown and need to be estimated using *sample data.*
- The equation of the regression line contains the sample y intercept, $\hat{\beta}_0$, and the sample slope, $\hat{\beta}_1$.

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

where: $\hat{y}$ = the predicted value of y
$\hat{\beta}_0$ = the sample y intercept
$\hat{\beta}_1$ = the sample slope

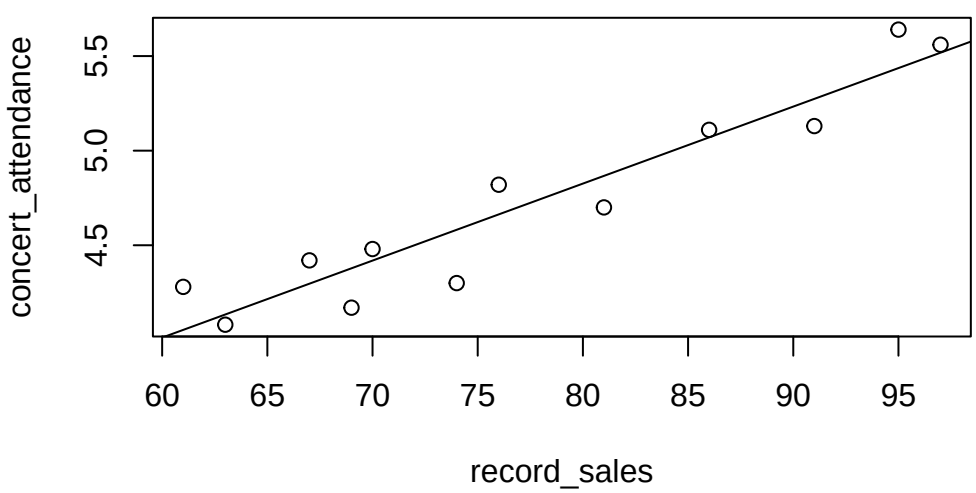
- To determine the equation of the regression line for a sample of data, the researcher must determine the values of $\hat{\beta}_0$ and $\hat{\beta}_1$.
- This process is sometimes referred to as **least squares analysis**.
- Least squares analysis is a process whereby a regression model is developed by *producing the minimum sum of the squared error values.*
- The least squares regression line is the **regression line that results in the smallest sum of errors squared**.

Example

The data in the table are the twelve observations corresponding to concert attendance (in thousand) and total worldwide CD sales by the performing artist (or band) in the previous year (also in thousand).

i. Draw a scatter diagram for the data and determine by inspection if there exists an approximate linear relationship between X and Y.

Number of CD sales ('000) x	Concert attendance ('000) y	x^2	xy	Predicted value $\hat{y}$	Residual $y - \hat{y}$
61	4.280				
63	4.080				
67	4.420				
69	4.170				
70	4.480				
74	4.300				
76	4.820				
81	4.700				
86	5.110				
91	5.130				
95	5.640				
97	5.560				




```
integer(0)
```

ii. Use the data to develop a regression model to predict concert attendance by CD sales.

4.2.2 Residual analysis

- Each difference between the actual y values and the predicted y values is the error of the regression line at a given point, $y - \hat{y}$, and is referred to as the **residual**.
- Except for rounding error, the sum of the residuals is approximately zero
- The analysis of residuals plays an important role in validating the regression model.
- Residual analysis plots are a very useful tool for assessing aspects of accuracy of a linear regression model on a particular dataset and testing that the attributes of a dataset meet the requirements for linear regression.
- The following are the assumptions of simple linear regression analysis

1. The model is linear.
2. The error terms have constant variances. (*The assumption of constant variance is called homoskedasticity. If the error variances are not constant, it is called heteroskedasticity.*)
3. The error terms are independent.
4. The error terms are normally distributed.

- Residual plots can be used to show if there are problems with the dataset and the model produced that need to be considered in looking at the validity of the model.

Residual plot

- A particular method for studying the behaviour of residuals is the residual plot.
- The residual plot is a type of graph in which the residuals for a particular regression model are plotted along with their associated value of x and an ordered pair $(x, y - \hat{y})$.
- Information about how well the regression assumptions are met by the particular regression model can be obtained by examining the plots.
- Residual plots are more meaningful with larger sample sizes.
- For small sample sizes, residual plot analyses can be problematic and subject to over-interpretation.

Normal Q–Q (quantile-quantile) Plot

- Residuals should be normally distributed.
- A normal quantile plot (also known as a quantile-quantile plot or QQ plot) is a graphical way of checking whether your data are normally distributed.
- If residuals follow close to a straight line on this plot, it is a good indication they are normally distributed.

4.2.3 Coefficient of determination

- A widely used measure of fit for regression, models is the **coefficient of determination** (R^2)

- It is a statistical measure of **how well the regression predictions approximate the real data points.**
- In statistical point of view, the coefficient of determination is the proportion of variability of the dependent variable (y) accounted for, or explained by, the independent variable (x)
- The coefficient of determination ranges from 0 to 1.
- An R^2 of zero means that the predictor accounts for none of the variability of the dependent variable and that there is no regression prediction of y by x.
- An R^2 of 1 means perfect prediction of y by x and that 100% of the variability of y is accounted for by x (*i.e.* an R^2 of 1 indicates that the regression predictions perfectly fit the data.)

4.2.4 Relationship between r and R^2

- Let r be the coefficient of correlation.
- The coefficient of determination (R^2) is the square of the coefficient of correlation

$$r = \sqrt{R^2}$$

- The researcher must examine the sign of the slope of the regression line to determine whether a positive or negative relationship exists between the variables and then assign the same sign to the correlation value.

4.2.5 Multiple Linear Regression

- Multiple regression is an extension of ordinary least-squares (OLS) regression that involves more than one explanatory variable.
- Multiple linear regression (MLR) is a statistical technique that uses several explanatory variables to predict the outcome of a response variable.
- The goal of multiple linear regression (MLR) is to model the **linear** relationship between the explanatory (independent) variables and response (dependent) variable.

General equation for the probabilistic multiple linear regression model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \cdots + \beta_k x_k + \epsilon$$

where

- y = the value of the dependent variable/ response variable
- β_0 = the regression constant, or intercept
- β_1 = the partial regression coefficient for independent variable 1
- β_2 = the partial regression coefficient for independent variable 2
- β_k = the partial regression coefficient for independent variable k
- k = the number of independent variables.
- The **partial regression coefficient of an independent variable**, β_j, represents *the increase that will occur in the value of y from a one-unit increase in that independent variable if all other variables are held constant.*
- The partial regression coefficients occur because more than one predictor is included in a model.
- the simplest multiple regression model is one constructed with two independent variables, where the highest power of either variable is 1. The regression model is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$$

The constant and coefficients are estimated from sample information resulting in the following model

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$$

- Simple regression models yield a line that is fit through data points in the xy plane.

- In multiple regression analysis, the resulting model produces a response surface.
- The procedure for determining formulas to solve for multiple regression coefficients is similar. the formulas are established to meet an objecitve of minising the sum of squares of error for the model.

  Example

  A real estate study was conducted in a major city to investigate the factors influencing the market price of residential homes. Data were collected on 23 houses, recording the market price (in thousands of dollars), the size of each house in square metres, and the age of each house in years. As a data scientist working for a real estate company, you are required to develop a multiple linear regression model to predict the market price of a house based on its size and age.

1. Using the given dataset, perform an exploratory data analysis to summarize and visualize the relationships among the variables.
2. Fit a multiple linear regression model with market price as the dependent variable and the other two variables as predictors.
3. Clearly interpret the regression coefficients, assess the statistical significance of the predictors, and evaluate the overall fit of the model.
4. Use your model to predict the price of a house that is 250 square metres in size and 12 years old, and discuss any assumptions made in the regression analysis.

References

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5 Summary

In summary, this book has no content whatsoever.

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References